



Territory Briefing: Modena

Professional province-level screening based on ARCUS historical evidence, territorial exposure, ARCUS/AINOP Collapse Rate and historical-pattern reading.

PRIORITY

53/100

COLLAPSE

1.61x / national rank 29

EVENTS

4

DECISION MESSAGE

Distributed chronic risk

REPORT DATE

Jun 10, 2026

CONTEXT

Bridge

PROVINCE

Modena

OUTPUT

New construction

ARCUS is built on the first systematic database of collapsed bridges in Italy from 2000 to today: the report does not merely rank cases, it reads documented historical patterns to guide preliminary checks.

The selected province requires preliminary screening for Bridge interventions. The value is not the case list: it is understanding whether Modena has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

RECOMMENDED USE

ARCUS reading: for new Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Modena.

PRIORITY INDEX

53/100

exposure index + heritage reading

COLLAPSE RATE

1.61x

ARCUS/AINOP national benchmark; confidence high

HISTORICAL EVENTS

4

Hydraulic dominant

RELIABILITY

85/100

8 linked sources

COLLAPSE RATE INTERPRETATION

The 1.61x rate versus the national average places Modena among the provinces with documented historical vulnerability above the ARCUS/AINOP benchmark. Modena is national rank 29 by documented ARCUS/AINOP collapse rate. Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.



MODENA / ARCUS ATLAS EXTRACT

GEOMORPHOLOGY AND CONTEXT

Modena shows a hydraulic profile that should be read by geomorphological setting, not as a uniform signal: 4 ARCUS events across 4 municipalities indicate that confined or torrential watercourses, valley crossings and lowland river reaches must be separated during verification.

MAP READING

The map locates 4 ARCUS cases across 4 municipalities and should be read with the historical pattern: Distributed chronic risk. Dominant date: 2020-12-06.

HISTORICAL PATTERN

Distributed chronic risk. historical risk distributed across multiple events and dates (25% on the most recurrent date).

DOMINANT TRIGGER

100%

Hydraulic

HAZARD

hydraulic

dominant territorial signal

READING

distributed risk

25% on the most recurrent date; read as distributed temporal risk plus local event clusters.

- Use the territorial exposure indicators to identify the dominant Hydraulic signal.
- Read the documented distributed historical pattern and use individual cases as map/appendix references.
- For hydraulic-triggered collapses in Modena, priority data includes: hydraulic model with TR100/TR200 flood scenarios, preliminary scour depth estimate at piers, riverbed/bathymetry evidence at crossings and debris-transport indicators where mountain, confined-channel or minor-torrent basins are involved.
- Differentiate checks by geomorphology in Modena: torrential, confined or steep watercourses require scour susceptibility and debris-transport checks; lowland river crossings require flood level verification, inundation duration and backwater sensitivity.
- Use CSV and GeoJSON exports for technical coordination.

DATA REQUEST PACKAGE

TR100/TR200 hydraulic model; preliminary scour depth; riverbed/bathymetry evidence; debris transport indicators where relevant.

COLLAPSE RATE CONFIDENCE NOTE

Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

The following list is a documentary reference, not an operational P1/P2/P3 ranking. The decision message remains the provincial historical pattern.

ID	Date	Municipality	Severity	Cause
B20.12.01	Dec 06, 2020	Pavullo nel Frignano	Total Collapse	Hydraulic
B10.10.01	Oct 31, 2010	Frassinoro	Total Collapse	Hydraulic
B13.03.01	Mar 23, 2013	Zocca	Total Collapse	Hydraulic
B09.12.01	Dec 31, 2009	Palagano	Total Collapse	Hydraulic

Layer	Meaning
Priority Index	Territorial exposure index: synthesizes dominant hazard, ARCUS case density, event concentration and evidence reliability. It does not directly measure historical bridge-stock vulnerability.
Collapse Rate	Ratio between ARCUS cases and AINOP counted bridges in the province. Value: 1.61x, national rank 29; it should be read separately from the Priority Index.
Why Separate	Priority Index indicates where the territory has elevated hazard exposure; Collapse Rate indicates where the bridge stock has shown historical vulnerability above or below the national average.
Historical Failure Context	Historical-pattern reading: concentrated event or distributed risk over time.
Hydraulic Cause Category	Hydraulic cause category: ARCUS uses 'hydraulic' as the documented trigger when sources confirm a flood or high-water event as the proximate cause. Mechanism-level classification - scour, bank erosion, debris impact, overtopping - is applied only where primary technical sources provide
Evidence Window	Observed period: ARCUS database of collapsed bridges in Italy from 2000 to today; close events in the same time window are read as one scenario, not decision duplicates.
AINOP Coverage	Collapse Rate uses AINOP counted bridges as provincial denominator. Confidence: high; local undercounting may inflate the ratio.
Evidence Classes	A = primary institutional/technical source; B = reliable technical or media source; C = generic documentary evidence; D = weak evidence to strengthen.